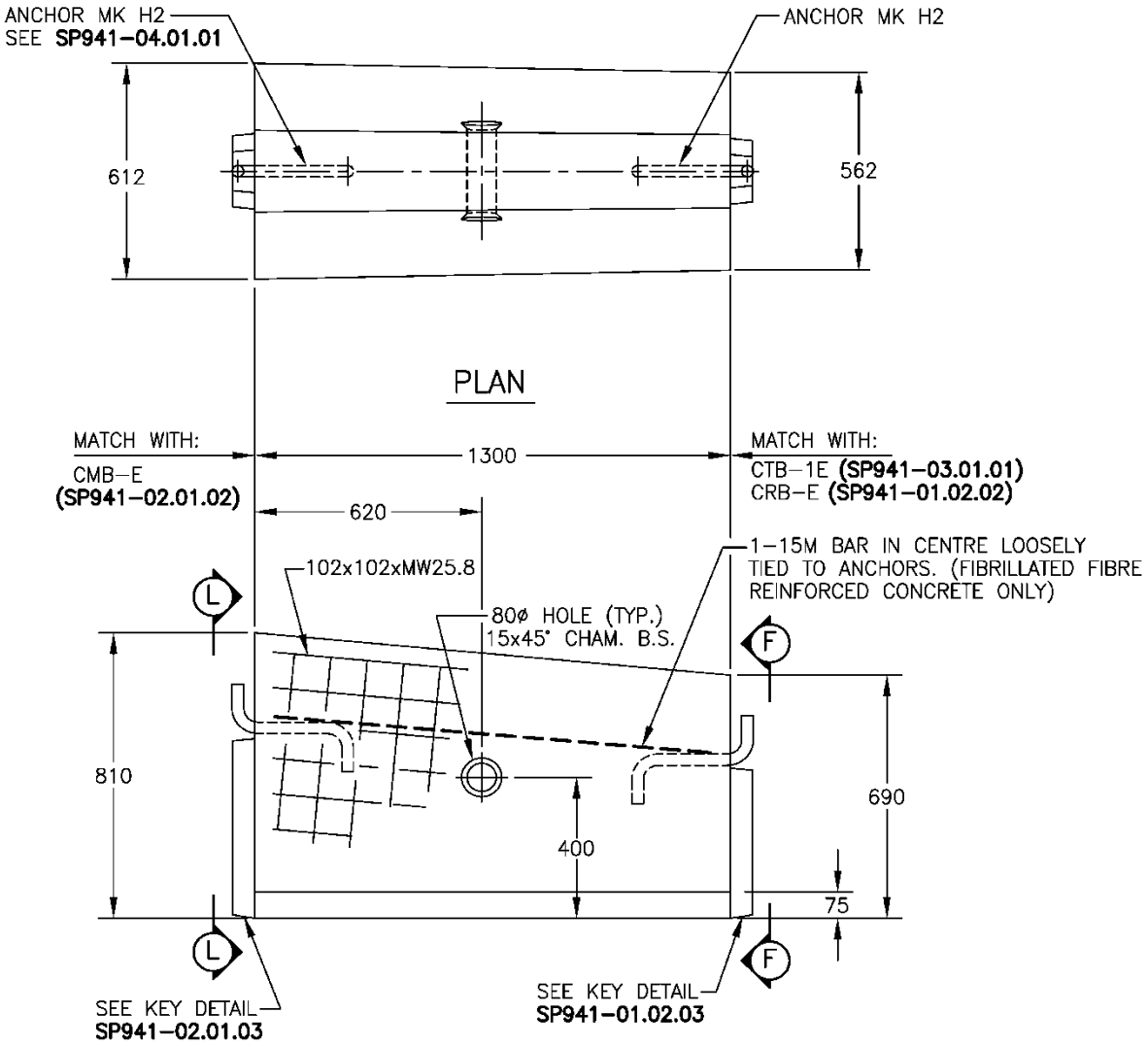
PRECAST CONCRETE TRANSITION BARRIER 810 mm TO 690 mm – CTB–2H	SP941–03.02.01

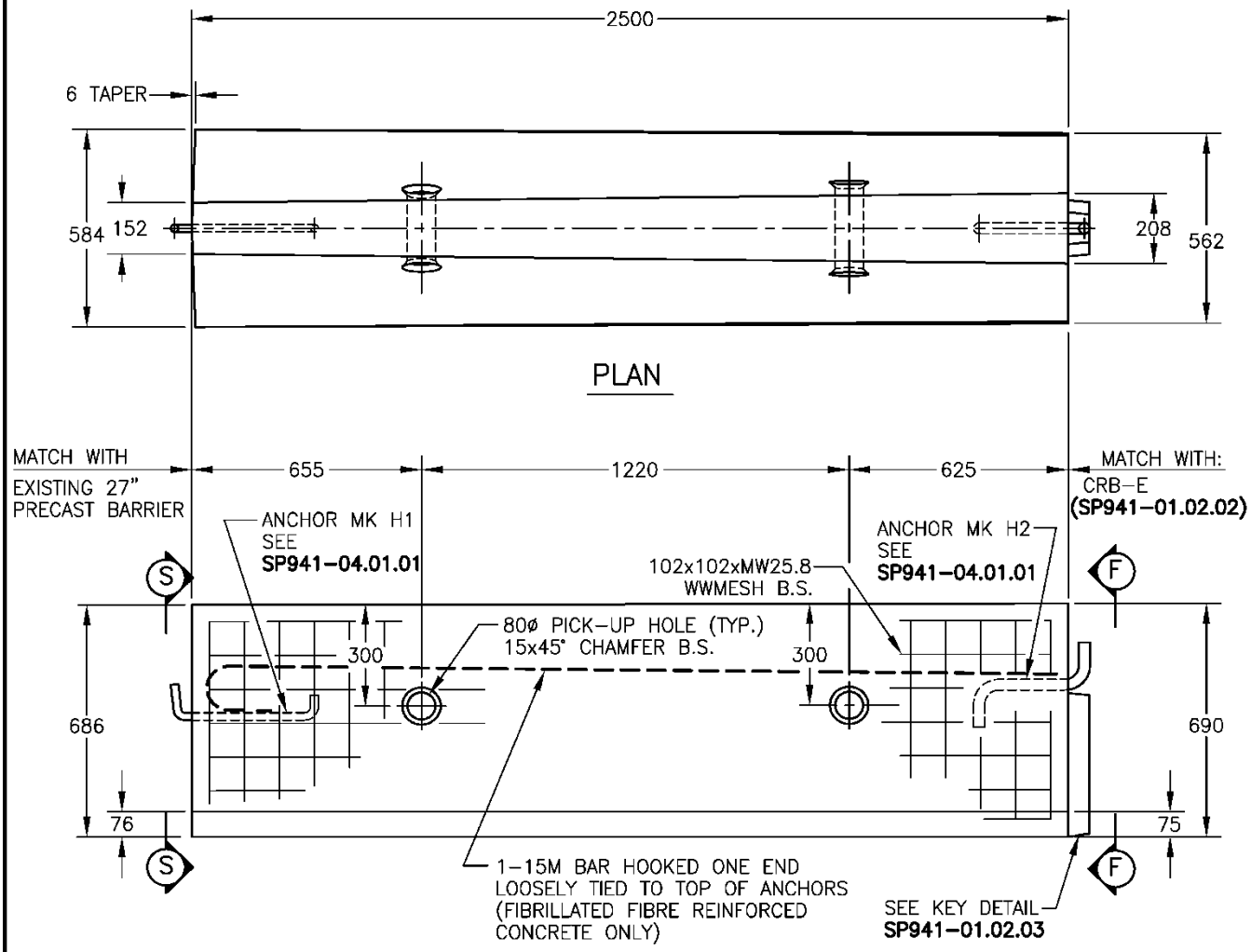


NOTES:

1. FOR VIEW (F) SEE **SP941–01.02.03.**
   FOR VIEW (L) SEE **SP941–02.01.03.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01.**
3. SEE DRAWING **SP941–02.01.02** FOR DETAILS OF 810 mm EYE UNIT.
   SEE DRAWING **SP941–01.02.02** FOR DETAILS OF 690 mm EYE UNIT
   OR **SP941–03.01.01** FOR DETAILS OF 690 TO 460 TRANSITION UNIT.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE TRANSITION BARRIER 686 mm TO 690 mm – CTB–3H	SP941–03.03.01



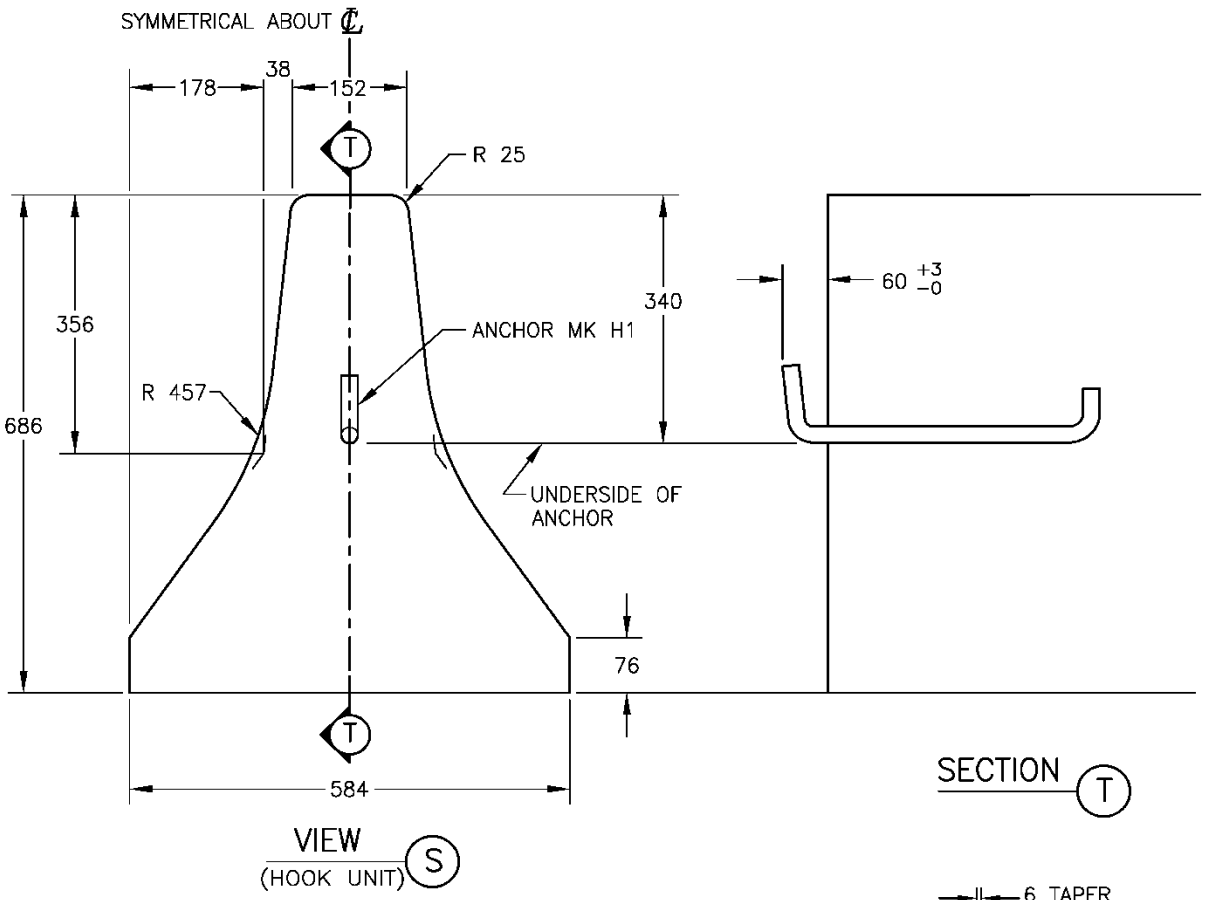
NOTES:

1. FOR VIEW (F) SEE **SP941–01.02.03.** FOR VIEW (S) SEE **SP941–03.03.02**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**
3. SEE DRAWING **SP941–01.02.02** FOR DETAILS OF 690 mm EYE UNIT.
4. THIS UNIT TO BE USED WITH EXISTING 27" BARRIER. 27" BARRIERS ARE NO LONGER USED IN NEW INSTALLATIONS

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

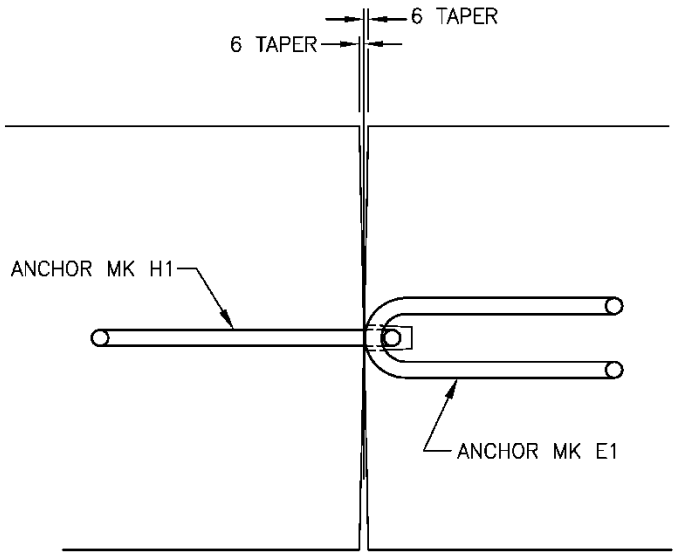
PRECAST CONCRETE TRANSITION BARRIER 686 mm TO 690 mm – CTB–3H DETAILS

SP941–03.03.02



VIEW (HOOK UNIT) S

SECTION T



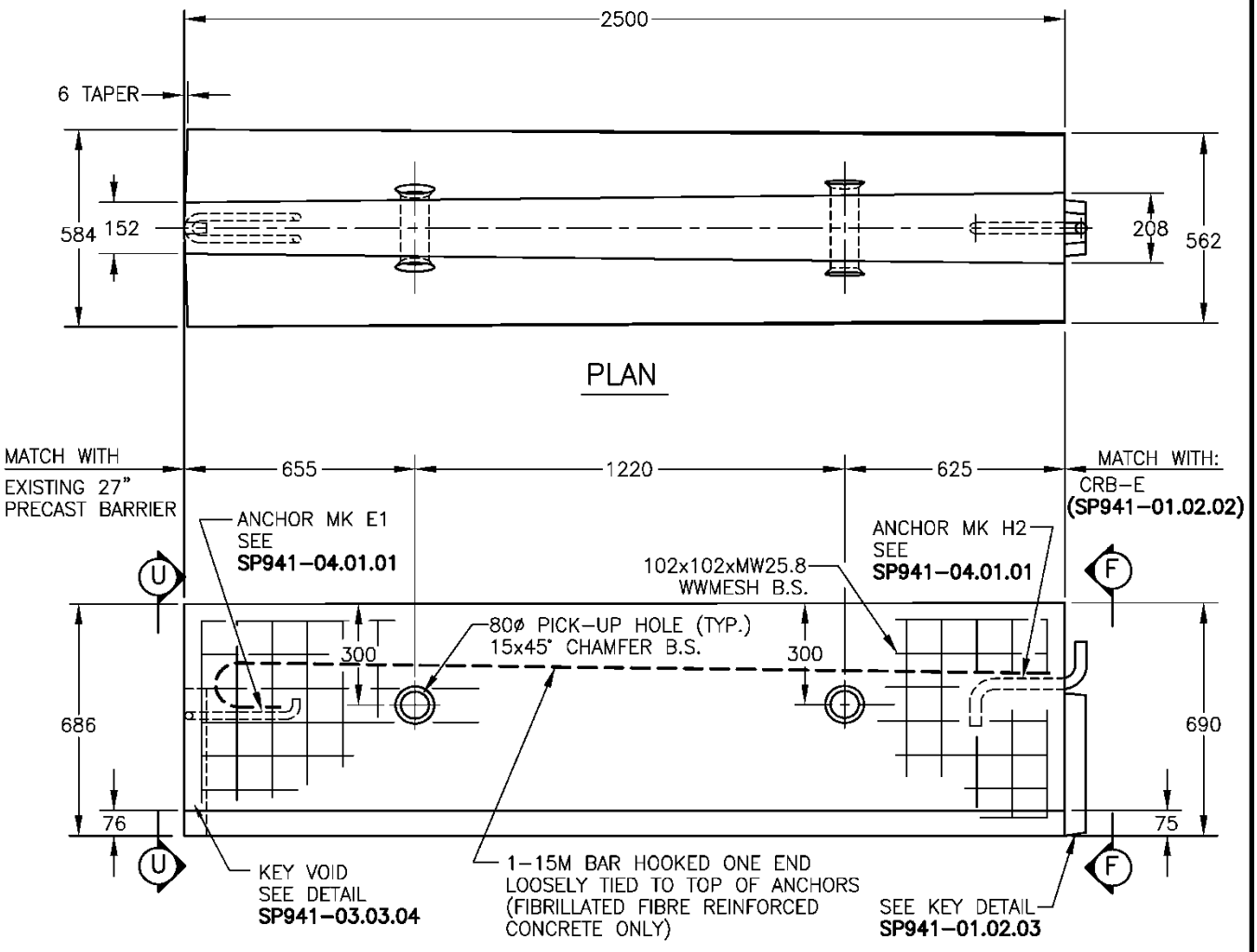
HOOK AND EYE DETAIL

NOTES:

1. FOR LOCATION OF VIEW (S) SEE **SP941–03.03.01**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**
3. FOR DETAILS OF ANCHORS SEE **SP941–04.01.01.**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE TRANSITION BARRIER 686 mm TO 690 mm – CTB–3E+H	SP941–03.03.03

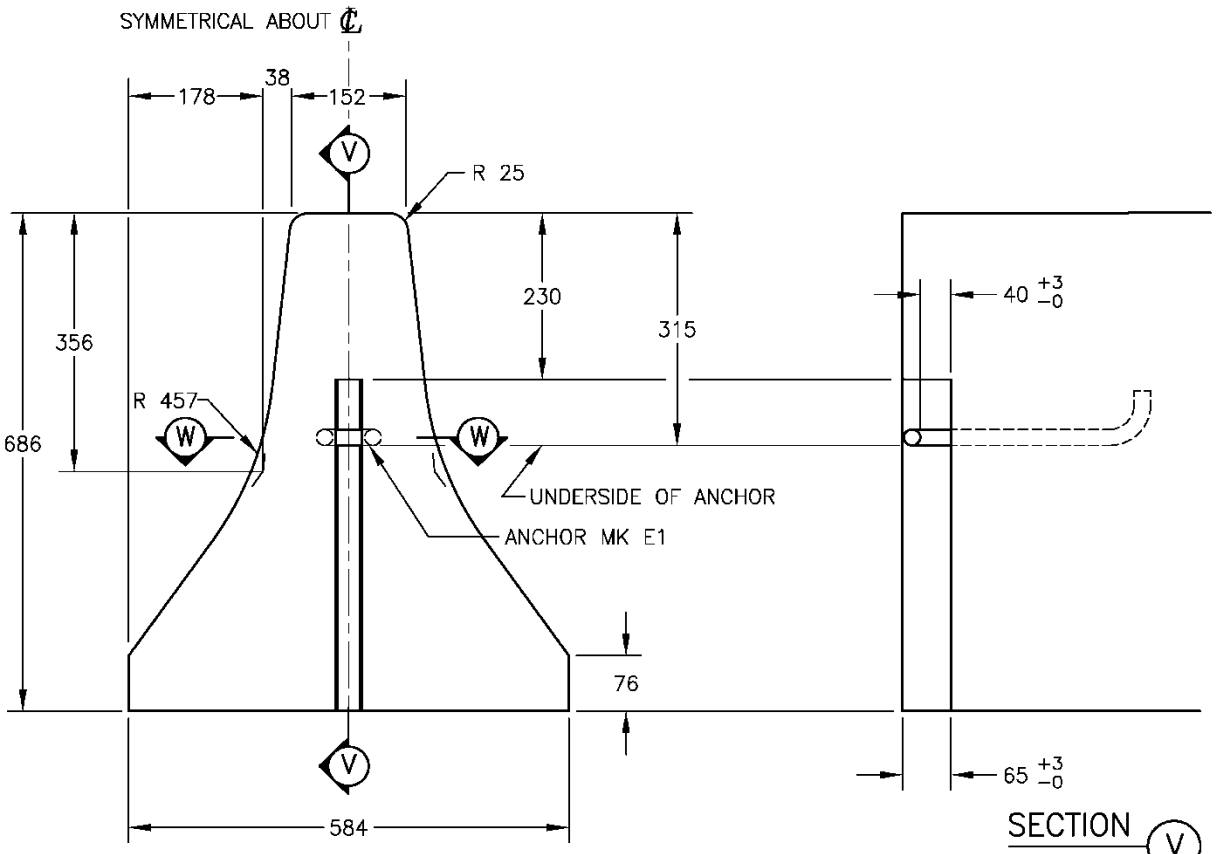


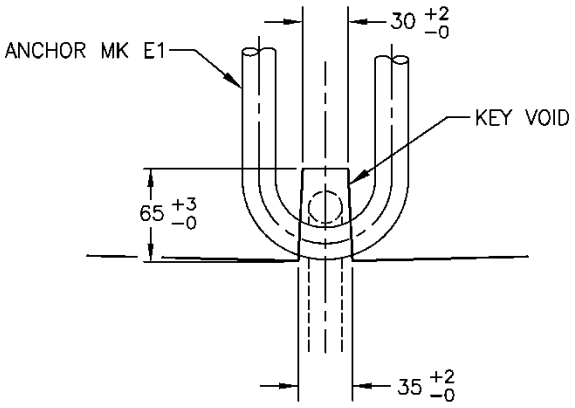
NOTES:

1. FOR VIEW (F) SEE **SP941–01.02.03.**. FOR VIEW (U) SEE **SP941–03.03.04**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**
3. SEE DRAWING **SP941–01.02.01** FOR DETAILS OF 690 mm HOOK UNIT.
4. THIS UNIT TO BE USED WITH EXISTING 27" BARRIER. 27" BARRIERS ARE NO LONGER USED IN NEW INSTALLATIONS

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE TRANSITION BARRIER 686 mm TO 690 mm – CTB–3E+H DETAILS	SP941–03.03.04





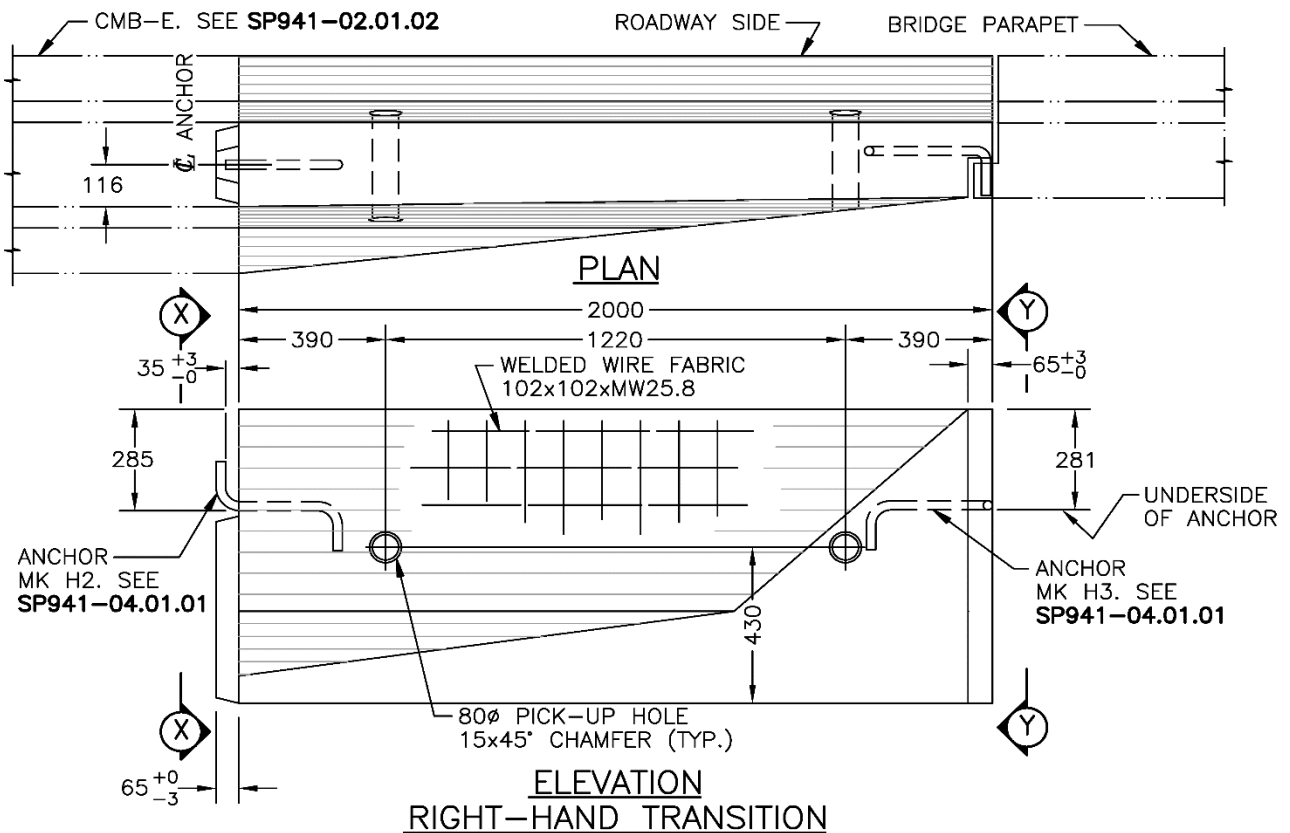
NOTES:

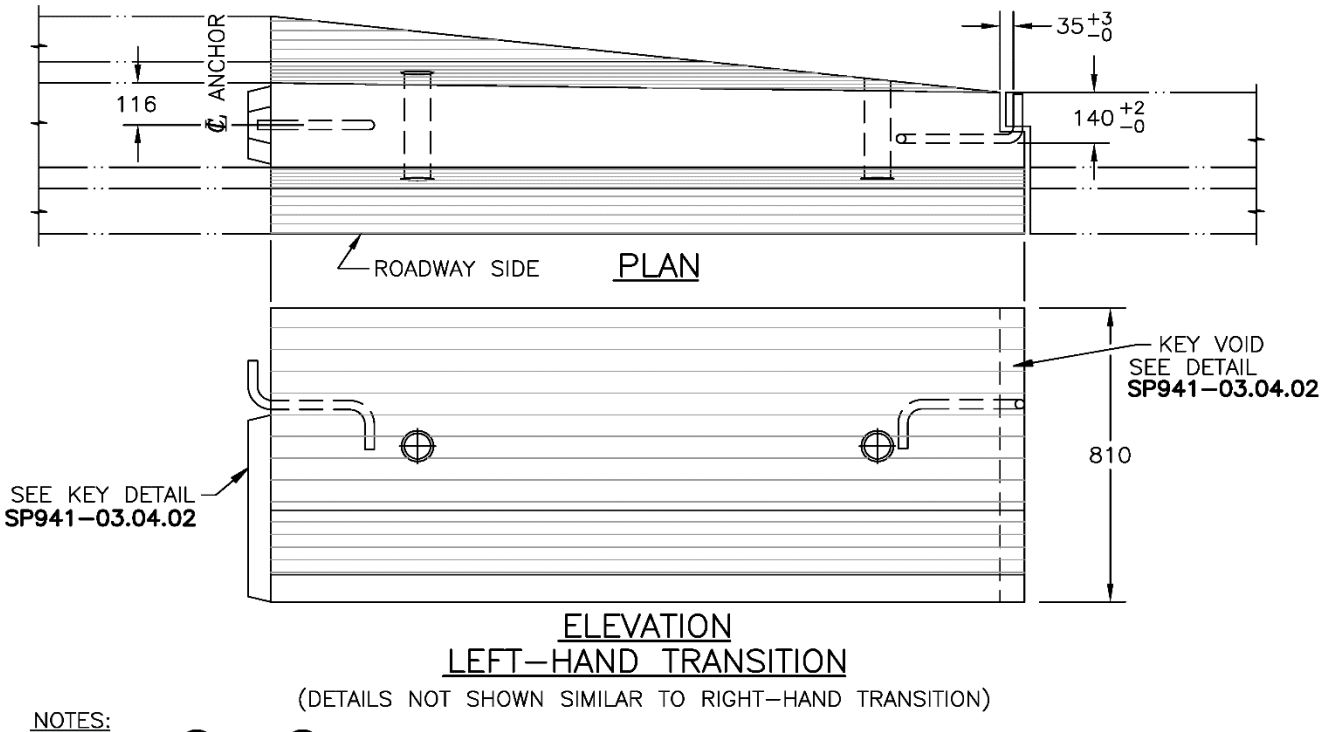
1. FOR LOCATION OF VIEW (U) SEE **SP941–03.03.03.**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**
3. FOR DETAILS OF ANCHORS SEE **SP941–04.01.01.**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

STANDARD BRIDGE PARAPET 810 mm HIGH TRANSITION

SP941–03.04.01





(DETAILS NOT SHOWN SIMILAR TO RIGHT–HAND TRANSITION)

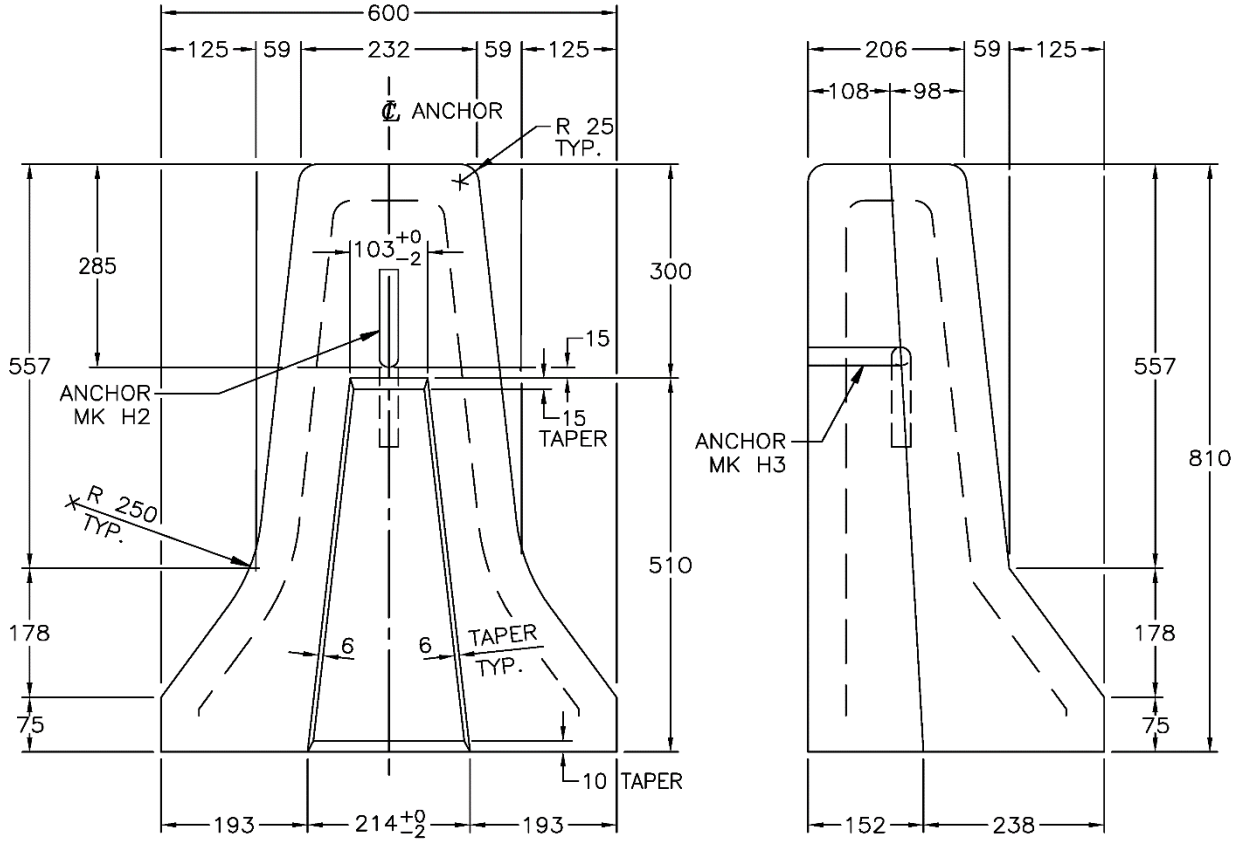
NOTES:

1. FOR VIEW (X) AND (Y) SEE **SP941–03.04.02**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

STANDARD BRIDGE PARAPET 810 mm HIGH TRANSITION – DETAILS

SP941–03.04.02

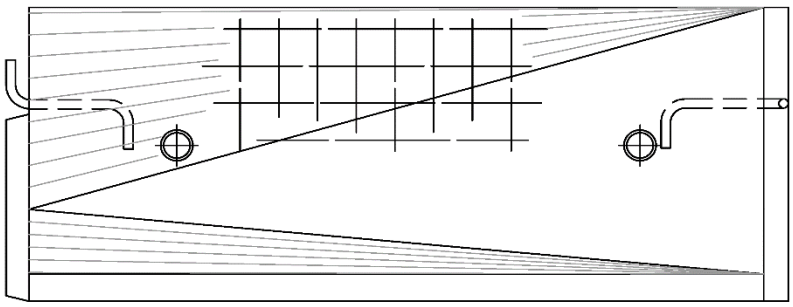


VIEW (X)

RIGHT HAND TRANSITION SHOWN
LEFT HAND TRANSITION SIMILAR

VIEW (Y)

RIGHT HAND TRANSITION SHOWN
LEFT HAND TRANSITION OPP. HAND



ACCEPTABLE ALTERNATE SIDE CONFIGURATION

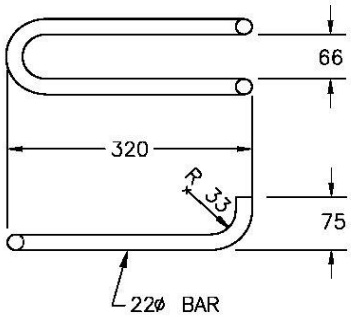
RIGHT HAND TRANSITION ELEVATION SHOWN
LEFT HAND TRANSITION ELEVATION OPPOSITE HAND

NOTES:

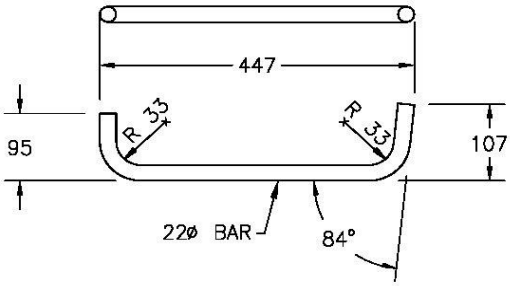
1. FOR LOCATION OF VIEW (X) AND (Y) SEE **SP941–03.04.01**
2. FOR GENERAL NOTES SEE **SP941–01.02.01**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

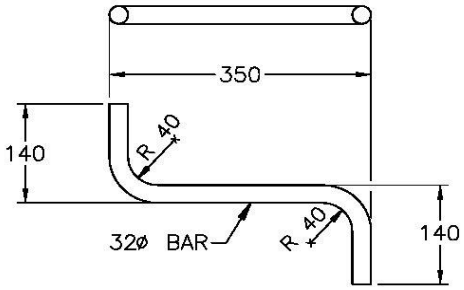
NO POST BARRIER ANCHORING HARDWARE	SP941–04.01.01



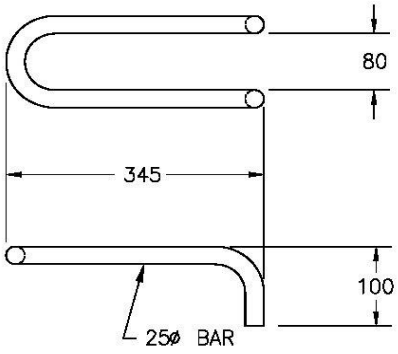
ANCHOR MK E1



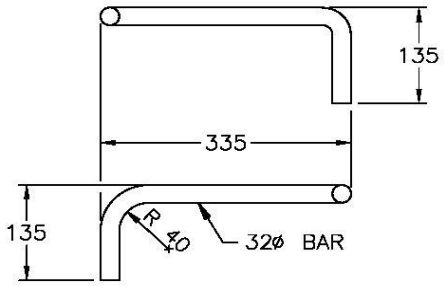
ANCHOR MK H1



ANCHOR MK H2



ANCHOR MK E2



ANCHOR MK H3

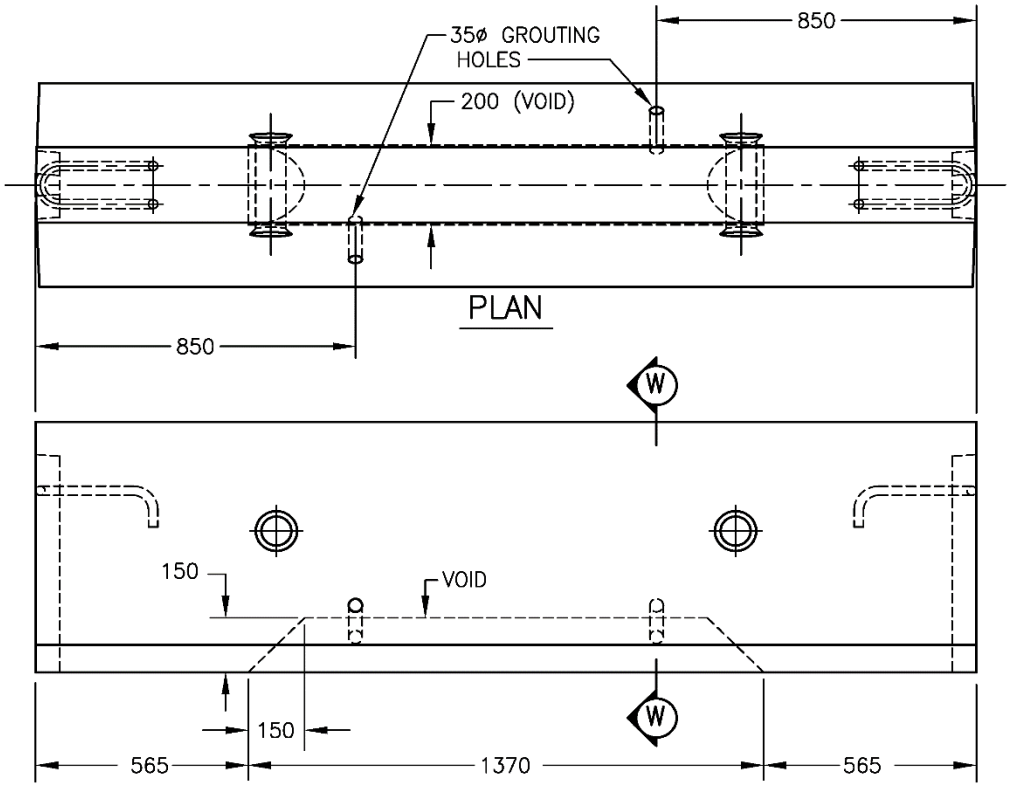
NOTES:

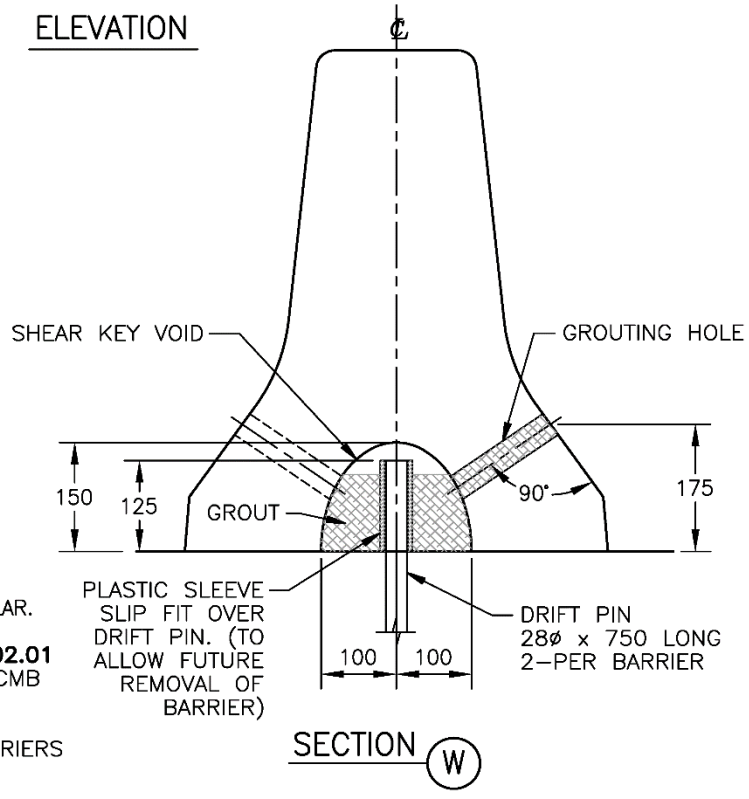
1. STEEL FOR ANCHORS TO CONFORM TO CSA SPECIFICATION CAN3–G40.21M.
2. PRIOR TO DELIVERY, EXPOSED SURFACES OF CONNECTIONS SHALL BE PREPARED FOR AND GIVEN A HEAVY APPLICATION OF ZINC RICH COATING SELECTED FROM THE MINISTRY'S RECOGNIZED PODUCTS LIST..

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

PRECAST CONCRETE ROADSIDE AND MEDIAN BARRIERS SHEAR KEY VOID DETAILS

SP941−04.02.01





NOTES:

1. 690 HIGH CRB AS SHOWN. 810 HIGH CMB SIMILAR.
2. FOR DETAILS OF 690 CRB SEE **SP941−01.02.01** & **SP941−01.02.02.** FOR DETAILS OF 810 CMB SEE **SP941−02.01.01 & SP941−02.01.02.**
3. DRAINAGE SLOTS ARE NOT ALLOWED IN BARRIERS WHEN SHEAR KEYS ARE REQUIRED.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 942

PRECAST CONCRETE INTERLOCKING MODULAR BLOCKS

942.01 Scope – This Section covers the quality and manufacture of precast concrete interlocking modular blocks (Interlocking Blocks) made using wet-cast concrete supplied for Ministry use.

This standard specification covers only Interlocking Blocks that have the following nominal dimensions:

(a) Full-sized blocks: 750 mm height x 750 mm depth x 1500 mm long.

(b) Half sized blocks: 750 mm height x 750 mm depth x 750 mm long.

(c) Compatible corner blocks and other partial sized or special purpose blocks for use with the full-sized blocks

Requirements for other block sizes, types and proprietary systems will be specified in the Contract.

Interlocking Blocks shall be supplied in the sizes and types as required by purchase order or Contract/Drawing requirements, and shall comply with this Section unless otherwise specified by the Contract or Purchase Order.

942.02 Definition of Precast Concrete Interlocking Modular Blocks ("Interlocking Blocks") – Precast monolithic concrete blocks with interlocking shear keys of specified dimensions made from wet-cast concrete for use in the construction of an engineered gravity retaining wall or an engineered mechanically stabilized earth (MSE) wall.

MATERIALS

942.10 Provision of Interlocking Blocks – Interlocking Blocks shall be supplied from the products listed on the Ministry's Recognized Products List (RPL).

The Recognized Product List is available on-line at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/recognized-products-list

942.11 Details of Interlocking Blocks – All block edges, except the shear key and corresponding socket, shall have a 40 ± 20 mm chamfer.

Dimension tolerances shall be ± 15 mm.

All Interlocking Blocks require the following identification information to be cast into the end of each block, in a manner, size and depth to be permanently legible:

(a) manufacturer's name or trademark,

(b) date of manufacture,

(c) concrete mix design identification, and

(d) form number.

942.12 Shear Keys – Interlocking Blocks shall have engineered interlocking shear keys with no more than 12 mm of design clearance around the shear key. The shear keys shall be cast integrally with the block and shall provide shear strength across the interface between courses of the Interlocking Blocks.

The shear key shall be designed to allow interlocking

(a) in a linear installation in running bond and stack bond layouts, and

(b) at ninety degrees (90°) corners

The interface shear strength between the courses of Interlocking Blocks and the shear key strength shall be adequate to resist earth pressure forces of ten metres (10 m) in height. Engineered documentation and test results that demonstrate the interface shear strength and shear key strength shall be submitted to the Ministry Representative for compliance review upon request.

Bench style and wall top course Interlocking Blocks shall have a flat top surface (i.e. without shear keys) with recessed galvanized lifting loops.

942.13 Lifting Loop Strands – Contractor shall ensure that Interlocking Blocks shall have galvanized steel lifting loops designed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of BC, and be designed so that the block can be lifted and positioned using only a standard hook or clevis. When requested, the Contractor shall provide the authenticated design record for the Ministry's review.

Lifting loops shall be fabricated from 12.5 mm or larger nominal diameter 7-wire zinc coated steel wire strand. Zinc coated 7-wire strand shall be in accordance with ASTM A475 or CSA G12 Class A zinc coating.

Lifting loops shall be anchored and embedded sufficiently into the block to safely enable lifting and handling of the blocks.

Lift points shall be designed and installed directly above the centre of mass of the block and in a manner that will allow them to remain in place and not interfere with full concrete-on-concrete bearing when blocks are stacked on top of each other.

Installation shall not require the lift points to be removed.

942.14 Concrete Quality – All concrete material and work shall be in accordance with SS 211 except as modified by this SS 942.

Material shall be new or virgin and contain no Recycled Concrete Aggregate Material (RCAM).

Concrete shall meet the requirements of Table 942-A.

Table 942-A: Concrete Requirements

Property	**Criteria**
Minimum compressive strength at 28 days	32 MPa
Nominal Maximum Size of Coarse Aggregate (NMSA)	20 mm or 28 mm
Air Content: • 20 mm NMSA • 28 mm NMSA	 5% to 8% 4% to 7%
Slump	120 ± 30 mm
Maximum w/cm ratio by mass	0.40
Curing Type	In accordance with class of exposure C-2

942.14.01 Sulphate Exposure – The Ministry shall notify the Contractor if S-1 and/or S-2 exposure conditions exist for the soils present on-Site. If no information is provided by the Ministry, S-2 shall be assumed.

For imported bedding and back fill materials, the aggregate supplier shall perform the sulphate exposure assessment of its aggregate in accordance with CSA A23.2-3B testing requirements.

The test results will be used to determine the site-specific exposure level as defined in Table 3 of CSA A23.1. Cementitious materials to be used shall be in accordance with Table 3 of CSA A23.1, and be tested using CSA A3004-C6 for portland / portland-limestone cements or A3004-C8 for blended hydraulic cements or MS / HS equivalent cements to meet the requirement for the exposure level in the table

942.15 Finished Surface of Concrete – Concrete for Interlocking Blocks shall be poured monolithically, with no cold joints. (i.e. continuously poured until the concrete is level with the top edge of the form.)

All visible surfaces shall be given a Class 2 finish, and any other surfaces with a different surface finish as indicated on the Drawings shall be finished in accordance with SS 211.17.

Consistency of the concrete finish for the Interlocking Blocks for the project shall be maintained by using the same concrete mix and the same type of form oil.

Interlocking Blocks shall be sound and free of cracks or other defects that interfere with the proper placement of the block. Interlocking Blocks having cracks or spalling that are continuous through any solid element of the block shall be rejected regardless of the width or length.

All sharp or protruding surfaces of the Interlocking Blocks shall be removed by careful rubbing or grinding.

942.15.01 Defects – Structurally defective or damaged Interlocking Blocks, as well as those with defects outside the limits specified in SS 942.15.01(a) shall be cause for rejection.

(a) Repair of Limited Defects – Subject to the approval of the Ministry Representative, non-structural defects, within the limits identified below, may be repaired:

(i) Surface defects where their total area is not in excess of 2% of the exposed surface area of the unit.

(ii) Open pockets not greater than 6 mm in diameter and not more in number than twenty (20) in any 300 mm x 300 mm area.

(b) Repair Products – Approved repairs shall be done using approved patching materials from the Recognized Products List for patching of isolated small holes, cavities and similar self-confining defects of the Interlocking Blocks in order to restore the durability of the block.

The patching material shall provide a reasonable colour match to the cured dry Interlocking Block, be well tooled in, be finished flush and smooth, and cured adequately, using water or a curing compound.

942.16 MSE Walls – MSE walls that contain Interlocking Blocks shall be in accordance with the Ministry's Supplement to CSA S6, and the design of the MSE walls, including connection design and/or modification to the Interlocking Blocks, shall be prepared by a professional engineer registered with the Association of Professional Engineers and Geoscientists of BC.

Interlocking Blocks for the MSE walls shall be manufactured in accordance with the engineered MSE Wall design.

942.17 Quality Control Requirements for Interlocking Blocks – The manufacturer of the Interlocking Blocks shall have a Quality Control Plan in place in order to ensure and document compliance with the Contract requirements. This plan shall be submitted to the Contractor and the Ministry Representative for compliance review upon request.

Manufacturers failing to meet the Contract requirements shall take immediate remedial steps to become compliant.

In addition to the quality control requirements of SS 211, the Contractor shall extract a set of two 100 mm diameter by 200 mm long cores from one block prior to delivery to the Site.

The cores shall be taken in the middle of the block along the back face so that the block may still be used if specifications are met. The cored holes shall be patched and cured using approved materials from the Recognized Products List.

For projects with more than 50 blocks, a minimum of one set of cores shall be extracted for every full or partial lot of 50 blocks supplied.

The Contractor shall ensure that the core test results are traceable to every other block cast from the same concrete batch and every other block in the 50-block lot. Only Blocks with traceable testing documentation will be accepted for use.

One of these cores will be tested for the compressive strength in accordance with CSA A23.2-14C and CSA A23.2-9C.

The second core shall be tested for air void analysis in accordance with ASTM C457.

The evaluation of the results from the testing of the cores shall be in accordance with CSA A23.1.

The Ministry may reject the whole lot of blocks represented by the initial or subsequent core testing if the core tests fail to meet the Contract Specifications.

942.18 Quality Assurance of Interlocking Blocks – The Ministry Representative may elect to carry out Quality Assurance of the manufacturer's fabrication facility and processes and/or the Interlocking Blocks.

The manufacturer's facility and the Interlocking Blocks shall be accessible to the Ministry Representative for inspection, testing, quality assurance and quality audit purposes at all times during manufacturing and shipping of Interlocking Blocks.

All testing, material, and Quality Control records shall be made available to the Ministry Representative upon request.

The Contractor shall notify the Ministry Representative a minimum of fourteen (14) days in advance of the date the Interlocking Blocks are to proposed to be shipped to allow the Ministry Representative an opportunity to inspect the Interlocking Blocks before delivery.

Patching and/or repair of defects may be inspected before shipment, or upon delivery, and the rejected unit(s) shall be replaced at no cost to the Ministry.

The Ministry Representative reserves the right to conduct any Quality Assurance testing deemed necessary to confirm Interlocking Blocks comply with the specifications before Interlocking Blocks are accepted for final payment.

The Ministry Representative reserves the right to reject any or all Interlocking Blocks based on visual and non-destructive methods.

PAYMENT

942.90 Payment – Payment for INTERLOCKING BLOCKS shall be at the applicable Unit Prices for the Item(s) listed in the Schedule of Approximate Quantities and Unit Prices and shall be full compensation for everything furnished and done including, but not limited to, equipment, labour, quality control, supply of forms and all materials, placing, vibrating and curing air entrained concrete, stripping, finishing, patching and repairs, general clean-up and delivery of all Interlocking Blocks at Site.

Rejected Interlocking Blocks shall be replaced at the Contractor's expense, including shipping charges and removal of rejected Interlocking Blocks from site and from constructed walls.

SECTION 952

CONTRACTOR SUPPLY ASPHALT AND PAVING MATERIALS FOR HIGHWAY USE

952.01 Scope – This Section describes the requirements for the different types and grades of asphalt and paving materials to be supplied by the Contractor for highway use as set out in the Special Provisions of the Contract.

The materials so described are generally classified as follows:

- Asphalt Cements
- Cutback Asphalts
- Asphalt Emulsions
- Tall Oil Pitch
- Emulsified Products

The term "Supplier" referred to in this Section shall mean the party or parties manufacturing and supplying the Contractor with materials covered under this section, but the Contractor remains responsible to the Ministry for the quality of the materials provided.

The Ministry's Recognized Product List referenced herein may be found online at:

https://www2.gov.bc.ca/assets/gov/driving-and-transportation/transportation-infrastructure/engineering-standards-and-guidelines/recognized-products-list/recognized_products_list.pdf?forcedownload=true

952.02 Quality Control Requirements

952.02.01 Quality Control Plan – The Supplier shall develop and maintain a proven quality control plan in accordance with the provisions of this Section and the Special Provisions. The plan shall ensure that adequate inspection coverage is maintained throughout the entire process of manufacture and shipping. All supplies processed or manufactured within the Supplier's plant or procured from any other source shall receive sufficient inspection to ensure conformance specifications and evidence of such inspections may be provided to the Ministry Representative upon request.

Supplies not conforming to contractual requirements shall not be offered for highway use until the deviations have been authorized by the Ministry Representative.

It shall be a responsibility of the Contractor to ensure that all materials supplied satisfy the related Environmental and Health and Safety Regulations.

Product discharged directly into tank cars or tank trucks from "in line blending" facilities will be sampled as required to ensure uniformity and to satisfy quality requirements.

952.02.02 Quality Control Procedure – The Supplier may be required to furnish the Contractor with an outline of the quality control procedures detailing the method of implementing the requirements of this Specification. This outline shall include the following operations:

- sampling and testing
- storage and handling
- shipping
- recording and reporting.

952.02.03 Technical Requirements – The Contractor shall ensure that the Supplier provides the following:

(a) Batching and Batch Analysis: Each batch of product covered by this Section shall be given a batch number and be sampled and tested in accordance with the applicable specification.

The batch analysis must show the specification requirements for the product tested, test method employed, and the pertinent test results, as listed in Appendix 952-A.

The batch analysis shall also include:

- Details of the raw materials and their dosages if engineered vegetable oils are uses as modifiers; and
- a written declaration that none of the substances listed in SS 952.05(f) have been added.

The batch analysis shall be certified by an appropriately qualified and authorized representative of the Supplier.

The certified batch analysis shall be dispatched to the Contractor with the first load of material delivered. Provided the batch from which shipments are drawn remains the same and provided that the batch number is shown on the Bill of Lading, no further batch analysis will be required for subsequent remaining shipments. If, during the course of delivery, it is necessary to draw material from a different batch or batches of product, the analysis of the replacement batch shall be sent out with the first load of material from each and every replacement batch.

When certified batch results are not yet available, the Contractor may elect, at its own risk, to use the uncertified material. If the material is subsequently determined to be non-compliant with the specifications or certified batch results are not provided, remediation of any Work using such material will be to the Contractor's account.

(b) **Viscosity Chart (Absolute)** shall include the initial load and for each subsequent batch. If in-line blending is utilized, a new viscosity chart shall be issued should any change in viscosity occur.

952.03 Sampling and Testing – The Supplier shall retain adequate sampling equipment, employ satisfactory sampling procedures, and maintain the sampling program and records. The Contractor shall obtain samples for quality assurance.

The Ministry has developed Technical Bulletin GM19001 *Liquid Asphalt Sampling and Testing for the Ministry's Quality Audit*, which may assist the Contractor in establishing its own quality assurance procedures. This Technical Bulletin is available at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/technical-bulletins/geotechnical-materials

Test methods for asphalt and paving materials shall conform to the standard ASTM tests listed in Table 952-A or AASHTO tests listed in Table 952-B, supplemented by such special tests as may be described elsewhere to cover special or experimental type materials.

On request by the Ministry Representative, the Contractor shall provide samples, from the Supplier, of the product in quantities, not less than 4 L, as may be required for independent testing purposes.

952.04 Delivery of Asphalt and Paving Materials – The Contractor must ensure the Supplier delivers asphalt and paving materials in good condition, of uniformity of product, for any special storage requirements and at correct temperature to the specified delivery point.

Adequate records of delivery shall be kept, and every bill of lading must show:

- Type of product
- Batch number
- Mass
- Relative density at 15°C
- P.G. or Pen Specification (information)

Trucks shall be equipped with a submerged sampling valve system installed generally as shown on SS Drawing SP952-01 for each compartment in which asphalt material is to be carried out.

Delivery temperature of the shipment shall meet the viscosity requirement shown in Table 952-C.

Asphalt Emulsions and emulsified products shall be kept at temperatures above 5°C.

952.05 Asphalt Cement – Asphalt cements shall be subject to the following conditions:

(a) all supplied asphalt cement shall be Group A;

(b) shall be products prepared by the refining of petroleum oils;

(c) shall be homogeneous, free from water, and shall not foam when heated to 175°C;

(d) shall conform to either the Pen Grade requirements specified in Table 952-D or the Performance Grade (PG) the requirements specified in Table 952-E (for Polymer Modified Asphalt only) and Table 952-G;

(e) shall not:

(i) be air blown or catalytically oxidized in any way,

(ii) contain any air blown or catalytically oxidized residues, or

(iii) include residues caused by artificial distillation of coal, coal tar, or paraffin.

(f) shall not be modified using any of the following additives (although trace amounts in the source asphalt cement are permitted):

(i) polyphosphoric acid,

(ii) atactic polypropylene; carbon black; polyisobutylene; polyisoprene; natural rubber; alkaline bases; insoluble particulates or fibres; heavy metals, including salts of iron, copper, manganese and/or cobalt; silicates; styrene-butadiene rubber (random copolymer latex); synthetic waxes (paraffin waxes, naphthenic waxes);

(iii) synthetic and saturated oils (including but not limited to the following: vegetable oils or modified vegetable oils; (paraffin oils, polyalphaolefins (PAO), lube oils, and re-refined lube oils.);

(iv) waste oils (including but not limited to the following: cracked residues, re-refined high vacuum distillate oils; tall oils, vacuum tower asphalt extenders; waste cooking oils, waste engine oils, waste engine oil residues).

(g) may be modified using engineered vegetable oils; and

(h) The asphalt cement Supplier shall also declare in writing on the certified batch analysis provided under SS 952.02.03(a):

(i) if engineered vegetable oils are used as modifiers, the raw materials used and their dosages; and

(ii) that they have not added any of the substances listed in SS 952.05(f) above.

Upon request, the Contractor shall forward a 4 L sample of the product to the Ministry Representative.

NOTE: Values for penetration and viscosity are point values only. Intermediate values shall be obtained by interpolation between the tabulated values on a straight-line basis. Minimum viscosity at 60°C for other penetrations can be read from the corresponding straight line on SS Drawing SP952-02.

952.06 Cutback Asphalts – Cutback asphalts shall consist essentially of petroleum derivatives and shall be substantially free from water and other impurities.

Cutback asphalts shall be of the type and grade described in Table 952-H to Table 952-K.

952.07 Asphalt Emulsions – Asphalt materials in the form of aqueous emulsions shall be of:

- Anionic Type
- Cationic Type
- High Float Type

952.08 Anionic Type Emulsion for Road Purposes – Anionic type of emulsion requirements are as follows:

- The asphalt emulsion shall be homogeneous.
- It shall show no separation of asphalt after thorough mixing within 30 days (crack filler 20 days) after delivery, provided separation has not been caused by freezing.
- The emulsion shall conform to the requirements listed in Table 952-L.

952.09 Cationic Type Emulsions for Road Purposes – Cationic type emulsions shall conform to the requirements listed in Table 952-M and shall be uniform throughout.

952.10 High Float Emulsified Asphalt Storage stability – High float emulsified asphalt shall show no separation of asphalt within 30 days after delivery and shall be homogeneous after thorough mixing. The emulsion shall conform to the requirements listed in Table 952-N.

952.11 Tall Oil Pitch (TOP) – Tall oil pitch, co-product of the pulp and paper industry, is a residual fraction from the crude tall oil vacuum distillation process, and is typically used by the Ministry as an asphalt extender, recycling, and anti-stripping agent.

It shall be a dark brown resinous and viscous material insoluble in water but miscible with asphalt.

Where required, TOP will be added to asphalt cement in a ratio defined by the mix design, usually between 5% – 10%.

The composition and properties of TOP depends on the origin and type of trees from which it is derived and the extraction technology. Only TOP products listed in the Ministry's *Recognized Product List* shall be permitted for use on Ministry paving projects.

The TOP shall conform to the requirements listed in Table 952-O and Table 952-P.

952.12 Emulsified Products – Emulsified products consist of emulsified Tall Oil Pitch and proprietary asphalt-based emulsions. The emulsified products are typically used by the Ministry, in paving operations, as prime coats and tack coats and replace organic solvent-based cutback asphalts for environmental reasons.

Only products listed in the Ministry's *Recognized Product List* shall be permitted for use on the Ministry paving projects.

Each batch of the product delivered to the job site shall be accompanied by the written statement from the Supplier confirming that the product formulation complies with the original, previously approved formulation.

Emulsified Tall Oil Pitch and asphalt-based emulsified products shall conform to the requirements listed in Table 952-Q and Table 952-R.

Tables follow, starting on the next page.

Table 952-A: Standard Test Methods – ASTM

ASTM TEST DESIGNATION	TITLE OF TEST: Standard Test Method for…
D5	Penetration of Bituminous Materials
D13	Ductility of Asphalt Materials
D36	Softening Point of Bitumen (Ring-and-Ball Apparatus)
D70	Density of Semi-Solid Asphalt Binder (Pycnometer Method)
D92	Flash and Fire Points by Cleveland Open Cup Tester
D95	Water in Petroleum Products and Bituminous Materials by Distillation
D113	Ductility of Asphalt Materials
D139	Float Test for Bituminous Materials
D243	Residue of Specified Penetration
D244	Standard Test Methods and Practices for Emulsified Asphalts
D402	Distillation of Cutback Asphalt
D803	Testing Tall Oil
D1310	Flash Point and Fire Point of Liquids by Tag Open-Cup Apparatus
D1754	Effects of Heat and Air on Asphaltic Materials (Thin-Film Oven Test)
D2042	Solubility of Asphalt Materials in Trichloroethylene
D2170	Kinematic Viscosity of Asphalts
D2171	Viscosity of Asphalts by Vacuum Capillary Viscometer
D2415	Ash in Coal Tar Pitch
D6084	Elastic Recovery of Bituminous Materials by Ductilometer
D7496	Viscosity of Emulsified Asphalt by Saybolt Furol Viscometer

Table 952-B: Standard Test Methods – AASHTO

AASHTO TEST DESIGNATION	TITLE OF TEST: Standard Method of Test for…
M 320	Standard Specification for Performance-Graded Asphalt Binder
R 29	Standard Practice for Grading or Verifying the Performance Grade (PG) of an Asphalt Binder
T 44	Solubility of Bituminous Materials
T 48	Flash Point of Asphalt Binder by Cleveland Open Cup
T 240	Effect of Heat and Air on a Moving Film of Asphalt Binder (Rolling Thin-Film Oven Test)
T 313	Determining the Flexural Creep Stiffness of Asphalt Binder Using the Bending Beam Rheometer (BBR)
T 315	Determining the Rheological Properties of Asphalt Binder Using a Dynamic Shear Rheometer (DSR)
T 316	Viscosity Determination of Asphalt Binder Using Rotational Viscometer
T 350	Multiple Stress Creep Recovery (MSCR) Test of Asphalt Binder Using a Dynamic Shear Rheometer (DSR)

Table 952-C: Viscosity Requirements for Shipment Delivery Temperature

CLASS OF MATERIAL	KINEMATIC VISCOSITY (mm^2/s)
Cutback Asphalt for Surface Spraying	100 – 200
Cutback Asphalt and Asphalt Cement for Plant Mixing	100 – 500
Tall Oil Pitch	20 – 40

Note: In no case shall asphalt materials or Tall Oil Pitch be received at a viscosity outside the specified viscosity range.

Table 952-D: Penetration Grades of Group A Asphalt Cement

REQUIREMENTS	PENETRATION GRADE ASPHALT CEMENT								ASTM TEST METHOD
	80 – 100		120 – 150		150 – 200		200 – 300		
	Min	Max	Min	Max	Min	Max	Min	Max	
Penetration at 25°C 100 g and 5 s, 0.1 mm	80	100	120	150	150	200	200	300	D5
Minimum Viscosity at 60˚C, Pa·s	150	115	91	70	70	50	50	31	D2171
Flash Point, °C Cleveland Open Cup	230	---	220		220	---	175	---	D92
Thin Film Oven Test (TFOT) % Loss in Mass	---	0.80		0.85	---	1.3	---	1.5	D1754
Penetration of Residue at 25°C 100 g, 5 s, 0.1 mm, % of Original Penetration	55	---	47		50	---	45	---	D5, after TFOT.
Solubility in Trichloroethylene % by Mass	99.5	---	99.5		99.5	---	99.5	---	D2042
Ductility, 25°C 5 cm/min, cm	100	---			100	---	---	---	D113
Ductility, 15°C 5 cm/min, cm	---	---			---	---	100	---	D113

Table 952-E: Performance Grades of Asphalt Cement

PERFORMANCE GRADE ASPHALT CEMENT	PG 64-YY	PG 58-YY	PG 52-YY	AASHTO TEST METHOD
Min Low Temperature PG-**YY** Requirements for Minimum Pavement Design Temperature °C **PG-YY** may be as specified in Special Provisions	**-22**	**-28**	**-34**	M 320, Table 1 R 29
Solubility in TCE min. %wt.	99.0	99.0	99.0	T 44
Flash, min. °C	230	230	230	T 48
DSRo, |G*|/sin δ, min, kPa @ T°C	1.00 @ 64	1.00 @ 58	1.00 @ 52	T 315
Mass Change RTFOT, max ± %wt.	1.00	1.00	1.00	T 240
DSRr, |G*|/sin δ, min, kPa @ T°C	2.20 @ 64	2.20 @ 58	2.20 @ 52	T 315
DSRp, |G*|/sin δ, max, kPa @ T°C	5000 @ 25	5000 @ 19	5000 @ 13	T 315
BBR, m-value, min, @ T°C	0.300 @ -12	0.300 @ -18	0.300 @ -24	T 313
BBR, S, max, MPa @ T°C	300 @ 12	300 @ -18	300 @ -24	T 313
Dynamic Viscosity, max, Pa·s	3.00	3.00	3.00	T 316

Table 952-F: Polymer Modified Asphalt Viscoelastic Properties

Polymer Modified Asphalt PG Grade	Minimum $R_{3.2@58°C}$ per AASHTO T 350*
58-24 64-28	25%
58-37, 58-40 64-34 70-28	40%
64-37 76-28	55%

* AASHTO T 350: Multiple Stress Creep Recovery (MSCR) Test of Asphalt Binder Using a Dynamic Shear Rheometer (DSR)

Table 952-G: Pen/Performance Grade Equivalence

	PEN Grade (As specified in Table 952-D)			
	80/100A	**120/150A**	**150/200A**	**200/300A**
Equivalent Performance Grade – PG As specified in Table 952-E Min. Pavement design Temperature °C	PG 64-22	PG 58-28	PG 58-28	PG 52-34

Table 952-H: Slow Curing Cutback Asphalt

REQUIREMENTS	GRADE					
	SC – 70		SC – 250		SC – 3000	
	Min	**Max**	**Min**	**Max**	**Min**	**Max**
Flash Point, °C	65	---	80	---	105	---
Kinematic Viscosity at 60°C, mm^2/s	70	140	250	500	3 000	6 000
Residue from Distillation to 360°C, volume % by difference	70	90	80	94	95	---
Distillation Residue, Kinematic Viscosity at 60°C, mm^2/s	400	7 000	800	10 000	4 000	35 000
ASPHALT RESIDUE:						
Residue of 100 Penetration, % by Mass	50	---	60	---	80	---
Ductility of 100 Penetration residue at 25°C, cm	100	---	100	---	100	---
Solubility in Trichloroethylene % by mass	99.0	---	99.0	---	99.0	---
Water, % by Mass or Volume	---	0.5	---	0.5	---	0.5

Table 952-I: Medium Curing Cutback Asphalt

REQUIREMENTS	GRADE			
	MC – 30		MC – 250	
	Min.	Max.	Min.	Max.
Flash Point, °C	38	---	65	---
Kinematic Viscosity at 60°C, mm^2/s	30	60	250	500
Distillation Test, % of Total Distillate to 360°C: to 225°C	---	25	---	10
Distillation Test, % of Total Distillate to 360°C: to 260°C	40	70	15	55
Distillation Test, % of Total Distillate to 360°C: to 315°C	75	93	60	87
Residue from Distillation to 360°C, volume % by difference	50	---	67	---
PROPERTIES OF RESIDUE FROM DISTILLATION:				
Penetration at 25°C, 100 g, 5 s, 0.1 mm	120	250	120	250
Ductility at 25°C, cm	100	---	100	---
Solubility in Trichloroethylene, % by mass	99.0	---	99.0	---
Water, %	---	0.2	---	0.2

Table 952-J: Rapid Curing Cutback Asphalt

REQUIREMENTS	GRADE					
	RC – 30		RC – 70		RC – 250	
	Min	Max	Min	Max	Min	Max
Flash Point, °C	---	---	---	---	27	---
Kinematic Viscosity at 60°C, mm^2/s	30	60	70	140	250	500
Distillation Test, % of Total Distillate to 360°C: to 190°C	15	---	10	---	---	---
Distillation Test, % of Total Distillate to 360°C: to 225°C	55	---	50	---	35	---
Distillation Test, % of Total Distillate to 360°C: to 260°C	75	---	70	---	60	---
Distillation Test, % of Total Distillate to 360°C: to 315°C	90	---	85	---	80	---
Residue from Distillation to 360°C, volume % by difference	50	---	55	---	65	---
TESTS ON RESIDUE FROM DISTILLATION:						
Penetration at 25°C, 100 g, 5 s, 0.1 mm	80	120	80	120	80	120
Ductility at 25°C, cm	100	---	100	---	100	---
Solubility in Trichloroethylene, % by mass	99.0	---	99.0	---	99.0	---
Water, %	---	0.2	---	0.2	---	0.2

Note: The material shall not foam when heated to the spraying and mixing temperature range recommended by the Canadian General Standards Board

Table 952-K: Cutback Asphalt Primer RM20

REQUIREMENTS	Min.	Max.
Kinematic Viscosity at 60°C, mm^2/s	20	35
Distillation Test, % of Total Distillate to 360°C: to 190°C	---	60
Distillation Test, % of Total Distillate to 360°C: to 225°C	40	---
Distillation Test, % of Total Distillate to 360°C: to 260°C	70	---
Distillation Test, % of Total Distillate to 360°C: to 315°C	85	---
Residue from Distillation to 360°C, volume % by difference	50	---
TEST ON RESIDUE FROM DISTILLATION:		
Penetration at 25°C, 100 g, 5 s, 0.1 mm	80	200
Ductility at 25°C, cm	100	---
Solubility in Trichloroethylene % by mass	99.0	---
Water, %	---	0.2

Table 952-L: Requirements for Anionic Type Emulsions

REQUIREMENT	TYPE OF EMULSION									
	RS – 1		RS – 2		MS – 2		SS – 1		CRACK FILLER	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
Saybolt Furol Viscosity, at 25°C, Furol seconds	20	100	---	---	100	---	20	100	50	200
Saybolt Furol Viscosity, at 50°C, Furol seconds	---	---	75	300	---	---	---	---	---	---
Residue by Distillation, %	57	---	62	---	62	---	57	---	65	75
Settlement, 1 Day, %	---	1.5	---	1.5	---	1.5	---	1.5	---	1.5
Demulsibility: 50 mL of 0.1 N $CaCl_2$, %	---	---	---	---	---	30	---	---	---	---
Demulsibility: 35 mL of 0.02 N $CaCl_2$, %	60	---	60	---	---	---	---	---	30	---
Sieve Test, % Retained on 1 mm	---	0.10	---	0.10	---	0.10	---	0.10	---	0.10
Cement Mixing Test, %	---	---	---	---	---	---	---	2.0	---	---
TESTS ON RESIDUE:										
Penetration at 25°C, 100 g, 5 s	100	200	100	200	100	200	100	200	60	100
Solubility in Trichloroethylene, %	97.5	---	97.5	---	97.5	---	97.5	---	97.5	---
Ductility at 25°C, cm	60	---	60	---	60	---	60	---	60	---

Table 952-M: Requirements for Cationic Type Emulsions

REQUIREMENT	TYPE OF EMULSION			
	CRS – 1K		CRS – 2K	
	Min	Max	Min	Max
Saybolt Furol Viscosity at 50°C, Furol seconds	30	125	174	400
% Residue by Distillation	62	---	68	---
Settlement 1 Day, %	---	1.5	---	1.5
Sieve Test, % Retained on 1 mm Mesh	---	0.1	---	0.1
Oil Portion of Distillate, % of Total Volume	0	3	0	3
Particle Charge	POSITIVE		POSITIVE	
TESTS ON RESIDUE				
Penetration at 25°C, 100 g, 5 s	100	250	100	150
Solubility in Trichloroethylene, % by mass	97.5	---	97.5	---
Ductility at 25°C, cm	60	---	65	---

Table 952-N: Requirements for High Float Emulsified Asphalts

REQUIREMENT	GRADE													
	HF-100S		HF-150P HF -100P		HF-150S		HF-250S		HF-350S		HF-500M		HF-1000M	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
Residue by Distillation, % By Mass	62	---	62		62	---	62	---	65	---	65	---	65	---
Oil Distillate, % By Volume	1	4	0.5	4	1	4	1	6	1.5	6	1	6	1	7
Saybolt Furol Viscosity, @ 50°C, Furol seconds	35	150	35	150	35	150	35	150	75	400	50	---	50	---
Sieve Test, % Retained on 1 mm Sieve	---	0.1	---	---	---	0.1	---	0.1	---	0.1	---	0.1	---	0.1
Coating Test, %	90	---			90	---	90	---	---	---	---	---	---	---
Settlement, 1 Day, % By Mass	---	1.5	---	---	---	1.5	---	1.5	---	1.5	---	1.5	---	1.5
Demulsibility: 50 mL 5.55 g/L $CaCl_2$, % By Mass	75	---	75		75	---	---	---	---	---	---	---	---	---
Workability @ 10°C	---	---	---	---	---	---	---	---	---	---	---	---	Pass	---
TESTS ON RESIDUE														
Penetration at 25°C, 100 g, 5 s	*	*	**	**	**	**	**	**	**	**	---	---	---	---
Viscosity at 60°C, Pa·s											8	20	2	8
Float Test at 60°C, s	1200	---	1200		1200	---	1200	---	1200	---	1200	---	1200	---
Solubility in Trichloroethylene, %	97.5	---			97.5	---	97.5	---	97.5	---	97.5	---	97.5	---
Ash Content, % by Weight				10										
Elastic Recovery, %at 10°C			50											

* See SS Drawing SP952-02

** See SS Drawing SP952-03 or SS Drawing SP952-05

Note For HF-150P, penetration tests shall be conducted on residue which has been distilled to 201°C ± 5°C

Table 952-O: Tall Oil Pitch Viscosity and Penetration Data

REQUIREMENTS	Min.	Max.
Results on Original Sample		
Absolute Viscosity, 60°C, Pa·s	1	2
Kinematic Viscosity, 135°C, mm^2/s	25	35
Penetration, 4°C/100g, 5 sec, 0.1 mm	150	250
Results after Thin Film Oven		
Loss in weight, %	---	0.65
Absolute Viscosity, 60°C, Pa·s	---	3
Kinematic Viscosity, 135°C, mm^2/s	---	60
Penetration, 4°C, 100 g, 5 sec, 0.1 mm	75	---

Table 952-P: Tall Oil Pitch Data

REQUIREMENTS	ASTM TEST METHOD	Min.	Max.
Softening Point, °C	D36	---	35
Flash Point, Cleveland Open Cup, °C	D92	250	---
Fire Point, Cleveland Open Cup, °C	D92	275	---
Boiling Point, °C		320	---
Specific Gravity		0.94	0.98
Vapour Pressure, mm Hg		---	1
pH		3.75	4.25
Wood Extractive, %		98	---
Ash, %	D803	---	0.8
Moisture, %	D803	---	0.1
Fatty Acids, %		7	9
Resin Acids, %		5	7
Unsaponifiables, %		39	44
Neutrals, %		42	46
Acid Number		20	30

Table 952-Q: Emulsified Tall Oil Pitch

REQUIREMENTS	Min.	Max.
Viscosity, Saybolt Furol, 25°C, Furol seconds	10	30
Residue by Distillation, % by weight	40	---
Oil Distillate, % by volume	---	0.1
Settlement 24 hrs, %	---	1.5
pH	6	8
Particle Charge	Negative	
Miscibility with Water	Pass	
Specific Gravity, 20°C	Approx. 1.0	
Boiling Point, °C	100	
Freezing Point, °C	0	
Vapour Pressure, mm Hg	20	
Odour and Appearance – distinctive resinous odour, light yellow colour		
TEST ON RESIDUE – Shall conform to the requirements for Tall Oil Pitch listed in		

Table 952-R: Emulsified Products

REQUIREMENTS	Min.	Max.
Viscosity, Saybolt Furol, 25°C, Furol seconds	10	60
Residue by Distillation, % by weight	40	---
Oil Distillate, % by Volume	---	5
Settlement 24 hrs, %	Pass	
Miscibility with Water	Pass	
TEST ON RESIDUE		
Penetration, 25°C, 100 g, 5 sec, 0.1 mm	40	200

APPENDIX 952-A

DATA TO BE PROVIDED FROM THE SUPPLIER

The following data is required from the Supplier.

PEN GRADED ASPHALT CEMENTS

Grade and Type (when required)

A. Original Asphalt

Anti-stripping agents (if any) – provide product name and dosage rate.

Density @ 15°C, kg/L

- Viscosity @ 135°C
- Viscosity @ 60°C
- Penetration @ 25°C, 100 g/5 s
- Solubility in Trichloroethylene
- Ductility @ 25°C
- Flash Point C.O.C.
- % Retained Penetration after T.F.O.T. @ 25°C

B. Residue After T.F.O.T.
(when Type A required)

- Penetration @ 25°C
- Penetration @ 10°C
- Penetration @ 4°C

NOTE: *Viscosity – Temperature chart also required for each batch.*

CUTBACK ASPHALTS

A. Cutback Asphalts as Supplied

- Flash Point, T.O.C. (C.O.C. above 79°C)
- Density @ 15°C, g/cm^3
- Viscosity @ 60°C, mm^2/s
- Distillate, % by volume of Total
  - Distillate to 360°C
  - Distillate to 190°C
  - Distillate to 225°C
  - Distillate to 260°C
  - Distillate to 315°C
- Residue from Distillation to 360°C
- Volume % by Difference

B. Properties of Residue

- Penetration @ 25°C
- Ductility @ 25°C
- Solubility in Trichloroethylene, %

C. Residue of 100 Penetration by Mass

- Ductility of 100 Penetration at 25°C, cm

PERFORMANCE GRADED CEMENTS

A. Original Asphalt

Anti-stripping agents (if any) – provide product name and dosage rate.

- Density @ 15°C, kg/L
- Viscosity @ 135°C
- Solubility in Trichloroethylene
- Flash Point
- Stiffness, DSRo
- Mass Change
- Stiffness, DSRr
- Stiffness, DSRp
- BBR m-value
- BBR Stiffness.

EMULSIONS

A. Original Emulsion

- Saybolt Furol Viscosity @ 25°C
- Saybolt Furol Viscosity @ 50°C
- % Residue by Distillation
- Settlement in 5 Days, %
- Oil Portion of Disillate
- % to Total Volume

B. Tests on Residue

- Float Test @ 60°C, s
- Penetration @ 25°C, 100 g/5 s
- Ductility @ 25°C